

Work and Practical Session: Machine Learning

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Session 04:

Unsupervised Learning – K-Means

We consider this dataset:

x1	x2	Class
1	10	1
1,5	2	2
1	6	1
2	1,5	2
2	10	1
3	2,5	2
3	6	1
4	2	2

1. Is this data linearly separable?
2. If it is possible, give a line that separates this data into two classes.
3. Apply the K-means algorithm on the input data (x1 and x2) assuming the initial centers are at (0,-4) and (0,8).
4. Let us define the **distortion function** J as: $J(c, \mu) = \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$, where m is the number of training examples, $x^{(i)}$ is the i -th training example, $c = \{c^{(1)}, c^{(2)}, \dots, c^{(m)}\}$ is the set of assigned classes for each training example, $\mu = \{\mu_1, \mu_2, \dots, \mu_k\}$ is the set of cluster centroids and k is the given fixed number of classes prior to executing K-means. Compute J when the initial centers are at (0,-4) and (0,8) for the given inputs, before applying the K-Means and for each iteration of the K-Means. What does this function J measure ?
5. Suppose you are given a 24-bit RGB image of size 256 by 256. How can you use K-means to compress this image into a 16-color image ? What is the compression rate of the obtained image ?